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Science Exam — 6th Grade · I Semester 2026

Chapter 1: Gravity and Other Forces

Date: March 19, 2026
Time: 7:15 am

Name: ______________________

Score: __________ / 35

Part I Mult. Choice / 10	Part II True/False / 5	Part III Vocab Match / 5	Part IV Calc. / 10	Part V Fundamentals / 5	TOTAL ___ / 35

■ **General Instructions:** Read each question carefully. You may use a basic calculator. Write your answers clearly.

Part I – Multiple Choice

Circle the letter of the best answer. (2 points each · Total: 10 pts)

1. What is mass?
- **a)** The force of gravity on an object
- **b)** The amount of matter in an object
- **c)** The speed at which an object moves
- **d)** The force that makes objects float

2. In which units is mass measured?
- **a)** Newtons (N)
- **b)** Kilometers (km)
- **c)** Grams (g) or kilograms (kg)
- **d)** Newton per kilogram (N/kg)

3. What is weight?
- **a)** The amount of matter in an object
- **b)** The speed of an object
- **c)** The downward pull force caused by gravity
- **d)** The upward force of water on an object

4. What is the gravitational force on Earth?
- **a)** 1.6 N/kg
- **b)** 3.7 N/kg
- **c)** 9.8 N/kg
- **d)** 23.1 N/kg

5. Which planet has the strongest gravitational force?
- **a)** Saturn
- **b)** Neptune
- **c)** Jupiter
- **d)** Earth

Part II – True or False

Write True or False. If False, correct the statement on the line. (1 point each · Total: 5 pts)

6. Mass changes depending on where you are in the solar system.
True ■ False ■

7. The Moon takes approximately 28 days to orbit Earth.
True ■ False ■

8. According to Einstein, gravity is a curvature in space-time.
True ■ False ■

9. An object floats when the buoyant force is less than its weight.
True ■ False ■

10. The Sun's gravity keeps all planets in orbit around it.
True ■ False ■

Part III – Vocabulary Match

Match each term in Column A with its correct definition in Column B. (1 point each · Total: 5 pts)

Column A — Terms	Answer	Column B — Definitions
11. Weight	________	a) A force between surfaces that slows moving objects
12. Normal Force	________	b) The path one object takes around another
13. Buoyancy	________	c) The upward force water exerts on objects
14. Orbit	________	d) The downward pull force caused by gravity — measured in Newtons
15. Friction	________	e) The upward force a surface exerts on a resting object

Part IV – Weight Calculations

Use the formula W = m × g. Show your work. (2 points each · Total: 10 pts)

16. A student has a mass of 50 kg. What is her weight on Earth?
Formula: $W = m \times g$, $g = 9.8$ N/kg
Work: ____
Answer: _____________________

17. An astronaut has a mass of 70 kg. What is his weight on the Moon?
Formula: $W = m \times g$, $g = 1.6$ N/kg
Work: ____
Answer: _____________________

18. A rock has a mass of 5 kg. What is its weight on Mars?
Formula: $W = m \times g$, $g = 3.7$ N/kg
Work: ____
Answer: _____________________

19. A robot weighs 300 N on Earth (g = 9.8 N/kg). What is its mass?
Formula: $m = W \div g$
Work: ____
Answer: _____________________

20. A 60 kg astronaut lands on Jupiter. What is her weight?

Formula: $W = m \times g$, $g = 23.1$ N/kg

Work: ____

Answer: _____________________

Part V – Fundamental Concepts

Answer each question in 1–2 complete sentences. (1 point each · Total: 5 pts)

21. What is the difference between mass and weight?

22. Explain how forces act on an object at rest on a table. Name both forces.

23. How does Newton's view of gravity differ from Einstein's?

24. When does an object float in water? Mention the two forces involved.

25. Name one natural satellite of Earth and one of the Sun. What keeps them in orbit?